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APPENDIX A

Representative gas properties

Representative gas properties have been used to illustrate the theory and, more importantly, they are necessary for the application of the DSMC demonstration programs. The relationship between the molecular models and the real gas is based primarily on the coefficient of viscosity, together with the power law of the variation of this coefficient with temperature. Table A1 is based on the data in Chapman and Cowling (1970).

Table A1 Properties under standard conditions (101,325 Pa and 0°C).

Gas	Symbol	Degrees of freedom ξ	Molecular mass $m \times 10^{27}$ kg	Viscosity coefficient $\mu \times 10^{5}$ Nsm^{-2}	Viscosity index ω
Hydrogen	H_2	5	3.34	0.845	0.67
Helium	He	3	6.65	1.865	0.66
Methane	CH_4	6.4	26.63	1.024	0.84
Ammonia	NH_3	6.25	28.27	0.923	1.1
Neon	Ne	3	33.5	2.975	0.66
Carbon monoxide	CO	5	46.5	1.635	0.73
Nitrogen	N_2	5	46.5	1.656	0.74
Nitric oxide	NO	5	49.88	1.774	0.79
Sea level air		5	48.1	1.719	0.77
Oxygen	O_2	5	53.12	1.919	0.77
Hydr'n chloride	HCl	5	61.4	1.328	1.0
Argon	Ar	3	66.3	2.117	0.81
Carbon dioxide	CO_2	6.7	73.1	1.380	0.93
Nitrous oxide	N_2O		73.1	1.351	0.94
Sulphur dioxide	SO_2		106.3	1.164	1.05
Chlorine	Cl_2	6.1	117.7	1.233	1.01
Krypton	Kr	3	139.1	2.328	0.8
Xenon	Xe	3	218.	2.107	0.85

Eqn (4.62) with $\alpha = 1$ then gives the following reference diameters for the VHS molecular model.

Table A2 VHS molecular diameters at 273 K.

Gas	Diameter ($d \times 10^{10}$m)	Gas	Diameter ($d \times 10^{10}$m)	Gas	Diameter ($d \times 10^{10}$m)
H_2	2.92	N_2	4.17	CO_2	5.62
He	2.33	NO	4.20	N_2O	5.71
CH_4	4.83	Air	4.19	SO_2	7.16
NH_3	5.94	O_2	4.07	Cl_2	6.98
Ne	2.77	HCl	5.76	Kr	4.76
CO	4.19	Ar	4.17	Xe	5.74

The VSS molecular model can reproduce the measured values of the coefficient of diffusion as well as the coefficient of viscosity. With regard to diffusion between unlike molecules, and noting that the diameter for the cross-collisions is not necessarily equal to the mean of the diameters of the individual species, the matching of diffusion and viscosity coefficients does not lead to a unique set of parameters. In fact, a cross-collision diameter could be chosen to achieve this match even for the VHS molecule. However, the requirement for consistency between the viscosity and self-diffusion coefficients in a simple gas does lead to a uniquely defined value of α for simple gases.

Eqn (3.76) can be used to write the diameter of a VSS molecule in terms of the self-diffusion coefficient D_{11} as

$$d = \left(\frac{3(\alpha+1)(kT_{\text{ref}})^{\omega}}{8\Gamma(7/2-\omega)(\pi m)^{\frac{1}{2}} n (D_{11})_{\text{ref}} E_t^{\omega-\frac{1}{2}}} \right)^{\frac{1}{2}}. \tag{A1}$$

This may be combined with eqn (3.74) to relate α directly to the Schmidt number (Sc), i.e.

$$(Sc) \equiv \frac{\mu_{\text{ref}}}{\rho (D_{11})_{\text{ref}}} = \frac{2+\alpha}{(3/5)(7-2\omega)\alpha}. \tag{A2}$$

For a hard sphere gas (Sc) reduces to 5/6, and this is the subject of much discussion in Chapman and Cowling (1970). Reliable data for the self-diffusion coefficient is available for many gases from experiments with isotopes. The values of α from this data are presented in Table A3. The change in α from the VHS value of unity also leads to a small change in the reference diameter from eqn (4.62), and this is also tabulated.

Table A3 VSS molecular parameters

Gas	$\rho D_{11}/\mu$	α	d (at 273 K) $\times 10^{10}$ m
H_2	1.37	1.35	2.88
He	1.32	1.26	2.30
CH_4	1.42	1.60	4.78
Ne	1.35	1.31	2.72
CO	1.42	1.49	4.12
N_2	1.34	1.36	4.11
O_2	1.35	1.40	4.01
HCl	1.33	1.59	5.59
Ar	1.33	1.40	4.11
CO_2	1.375	1.61	5.54
Kr	1.29	1.32	4.70
Xe	1.33	1.44	5.65

The above values of α are for temperatures of the order of the standard temperature. Chapman and Cowling (1970) report data on Kr at 473 K and Xe at 378 K, and the corresponding values for α are 1.29 and 1.40, respectively. Data is also given for Ne, N_2, and O_2 at 78 K. The value of α for Ne is unchanged, but those for N_2 and O_2 increase to 1.52 and 1.45, respectively. Given the restricted temperature range and the uncertain accuracy of the data, it is not clear whether α can reasonably be regarded as a constant in applications that involve large changes in temperature.

Data is also available in Chapman and Cowling (1970) for the coefficient of diffusion D_{12} between a large number of gas pairs and, for some of these, the temperature exponent ω_{12} is also available. This data defines the reference values of these quantities, and the reference diameter $(d_{12})_{\text{ref}}$ will be assumed to be given by the mean value, as in eqn (1.35). Eqns (3.75) and (4.61) may then be combined to give the appropriate value of the VSS scattering parameter α_{12} as

$$\alpha_{12} = \frac{8(5-2\omega_{12})n(D_{12})_{\text{ref}}\pi(d_{12})_{\text{ref}}^2}{3(2\pi k T_{\text{ref}}/m_r)^{1/2}} - 1 . \tag{A3}$$

The resulting values of the VSS scattering parameter are presented in Table A4. For the cases in which a value of ω_{12} is not available, this parameter may be set to the mean value of ω for the two species. It should be noted that these are not the only sets of parameters that reproduce the transport properties.

Table A4 VSS parameters for cross-collisions.

Pair	α_{12}	ω_{12}	Pair	α_{12}	ω_{12}
H_2–He	1.20		Ne–CO_2	1.97	0.83
H_2–CH_4	1.54		Ne–Kr	1.59	
H_2–CO	1.33		Ne–Xe	1.84	
H_2–N_2	1.41		CO–N_2	1.54	
H_2–O_2	1.32	0.78	CO–O_2	1.54	0.71
H_2–Ar	1.47		CO–CO_2	1.61	
H_2–CO_2	1.69	0.84	N_2–O_2	1.38	
He–Ne	1.28		N_2–Ar	1.33	
He–N_2	1.50	0.69	N_2–CO_2	1.87	0.75
He–O_2	1.49		N_2–Xe	1.42	
He–Ar	1.64	0.725	O_2–Ar	1.33	
He–CO_2	2.13	0.84	O_2–CO_2	1.72	
He–Kr	1.81		O_2–Xe	1.34	
He–Xe	2.20		Ar–CO_2	1.63	0.805
CH_4–O_2	1.47	0.79	Ar–Kr	1.41	
CH_4–CO_2	1.67		Ar–Xe	1.47	
CH_4–Ar	1.46		Kr–Xe	1.44	

The DSMC calculations that are presented in §12.2 to §12.7 verify that the reference values of the VHS and VSS parameters that are listed in Tables A1 – A4 reproduce the transport properties of real gases at normal temperatures. The situation is less satisfactory for problems that involve very high temperatures in that reliable experimental values are not available.

Semi-empirical values of the viscosity of the components of air at high temperature have been given by Gupta et al. (1990). Their expression for molecular nitrogen is

$$\mu = 0.1 \exp(-11.8153)\, T^{0.0203 \ln(T) + 0.4329}. \qquad \text{(A4)}$$

This result is compared in Fig. A1 with that of Table A1. While the latter result is based on experiments at temperatures of the order of the standard temperature, its extrapolation to very high temperatures yields results that agree with eqn (A4) to within the uncertainty that is associated with this equation. The high-temperature calculations in this book have employed the low-temperature values, but the possible errors in these values at high temperatures should be kept in mind.

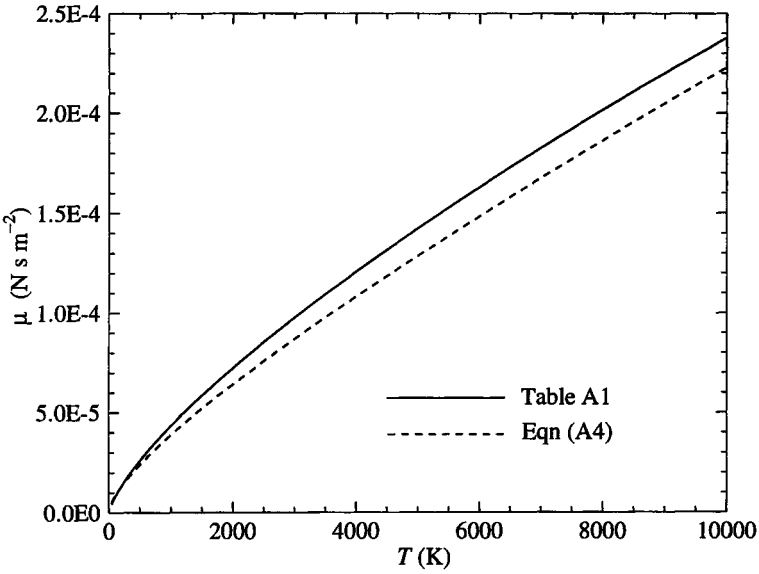


Fig. A1 Alternative values for the viscosity coefficient of molecular nitrogen at high temperatures.

The characteristic temperatures of rotation, vibration, dissociation, and ionization of typical diatomic molecules are listed in Table A5.

Table A5 Characteristic temperatures of diatomic molecules

Gas	Rotation Θ_r (K)	Vibration Θ_v (K)	Dissociation Θ_d (K)	Ionization Θ_i (K)
CH	20.4	4,102	40,300	129,000
Cl_2	0.35	801	28,700	
CO	2.77	3,103	29,700	162,000
H_2	85.4	6,159	52,000	179,000
N_2	2.88	3,371	113,500	181,000
NO	2.44	2,719	75,500	108,000
O_2	2.07	2,256	59,500	142,000
OH	26.6	5,360	50,900	

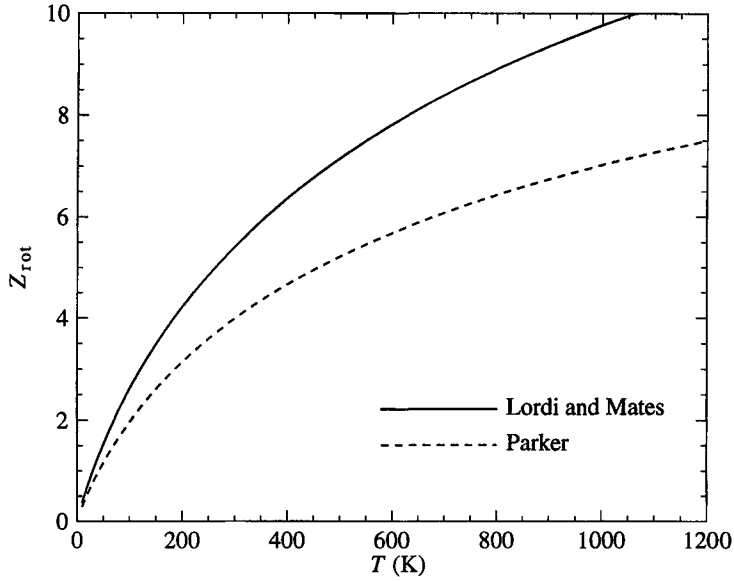


Fig. A2 Rotational relaxation collision number for air.

As noted in §1.7, rotation can generally be regarded as being fully excited, but data is needed for the rotational relaxation collision number Z_{rot}. Values have been inferred from a variety of experiments and, while these are subject to a considerable degree of scatter (Lumpkin et al. 1989), they are generally consistent with predictions based on the Landau–Teller model. Parker (1959) employed the equation

$$Z_{rot} = \frac{Z_{rot}^{\infty}}{1 + (\pi^{3/2}/2)(T^*/T_{tr})^{1/2} + (\pi + \pi^2/4)(T^*/T_{tr})} \qquad \text{(A5)}$$

with $Z_{rot}^{\infty} = 15.7$ and $T^* = 80.0$ When the data of Lordi and Mates (1970) is fitted to this equation the values are $Z_{rot}^{\infty} = 23.0$ and $T^* = 91.5$. The results from the two sets of coefficients are plotted in Fig. A2 for temperatures up to 1200 K. Little experimental data is available at higher temperatures, but Lumpkin et al. have proposed a model for nitrogen that predicts a reduction in Z_{rot} at higher temperatures. The values in this model depend on the ratio of the rotational to translational temperature as well as the overall temperature. The demonstration programs make provision for Z_{rot} to be specified as a second-order polynomial in the overall temperature. This is adequate for the approximation of eqn (A5) over a restricted temperature range, but would not cope with the more complex model in the context of a hypersonic flow calculation. Modified coding would be required and, since Z_{rot} is used in the collision subroutine, a table look-up would be preferable to repeated evaluation of the equation.

The vibrational relaxation rate is discussed in §6.8 where it was shown that the Millikan and White (1963) expression for the vibrational collision number can be put into the form

$$Z_v = (C_1/T^{\omega}) \exp(C_2 T^{-1/3}), \tag{A6}$$

where C_1 and C_2 are constants. Table A6 presents tentative values of these constants based on the data of Millikan and White. The values are for molecule 1 in a collision with 'molecule' 2. Note that values of C_1 and C_2 that are individually very different can lead to similar results for Z_v as a function of temperature.

Table A6 Constants for the evaluation of the vibrational collision number.

Molecule 1	Molecule 2	C_1	C_2
N_2		9.1	220.0
O_2		56.5	153.5
F_2		1007.0	65.0
CO		37.7	175.0
CO	Ar	7.27	221.
CO	He	1200.0	97.0
CO	H_2	2100.0	68.7
O_2	Ar	50.6	163.0
O_2	He	1950.0	65.8
O_2	H_2	4070.0	40.4
Cl_2		924.0	57.9
I_2		1750.0	29.0

The vibrational excitation program DSMC0V employs the levels that are given by the harmonic oscillator model and the characteristic vibrational temperature is the only data that is required. On the other hand, program DSMC0D employs anharmonic levels and the test case made use of the comprehensive data that has been assembled by Herzberg (1950). This is semi-empirical and the levels for nitrogen were expressed as eqn (11.35). The first coefficient is the characteristic temperature and the value of this is slightly different from that in Table A5. Variations of this order are generally present in data from different sources. The distribution of vibrational levels in other molecules are readily obtained from the data of Herzberg.

The chemical reaction rate theory of §6.6 and the comparisons of the dissociation–recombination results from program DSMC0D with the equilibrium composition required data on the partition function that are defined in eqns (6.31) – (6.34). The translational function depends only on the temperature, while the rotational and vibrational functions require the characteristic temperatures that are listed in Table A5. This electronic function of eqn (6.34) requires additional data and values for the degeneracies and characteristic temperatures are implicit in the following expressions that have been listed by Hansen (1976). Additional terms become progressively significant as the temperature increases, and these values are recommended for temperatures below 20,000 K in the case of atoms and for temperatures below 10,000 K for molecules.

$$Q_{el}(\mathrm{H}) = 2$$

$$Q_{el}(\mathrm{N}) = 4 + 10\exp(-27658/T) + 6\exp(-41495/T)$$

$$Q_{el}(\mathrm{O}) = 5 + 3\exp(-228.9/T) + \exp(-325.9/T) + 5\exp(-22830/T) + \exp(-48621/T)$$

$$Q_{el}(\mathrm{N_2}) = 1$$

$$Q_{el}(\mathrm{O_2}) = 3 + 2\exp(-11393/T) + \exp(-18985/T)$$

$$Q_{el}(\mathrm{NO}) = 2 + 2\exp(-174.2/T)$$

$$Q_{el}(\mathrm{H_2}) = 1$$

$$Q_{el}(\mathrm{Cl_2}) = 1 + 6\exp(-26345/T)$$

This Appendix is not intended to be a handbook for the best available data on the fullest possible range of gases. It merely provides reasonable values, for some common gases, of the physical quantities that are required for the application of the DSMC method. Alternative DSMC procedures would require different data. For example, the 'equilibrium collision theory' of §6.5 would require data on the chemical rate constants.

The values of the physical constants are included in the list of symbols.

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